Exp No: 2 Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

AIM:

To upload and analyze the data set given in csv format and perform data preprocessing and visualization.

PROGRAM:

```
from google.colab import drive   #mounting csv file to google colab
drive.mount('/content/drive')
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
df=pd.read_csv('/content/drive/MyDrive/SALES DATA.csv')
print(df.head()) #print the first few rows of data
print(df.isnull().sum())
df['Sales']=df['Sales'].fillna(df['Sales'].mean())
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
print(df.describe())
prdt_summary=df.groupby('Product').agg({
   'Sales':'sum',
   'Quantity':'sum'
}).reset_index()
print(prdt_summary)
plt.figure(figsize=(10, 6))
plt.bar(prdt_summary['Product'], prdt_summary['Sales'])
plt.title('TOTAL SALES BY PRODUCT')
```

```
plt.xlabel('PRODUCT')
plt.ylabel('TOTAL SALES')
plt.show()
plt.figure(figsize=(10, 6))
plt.bar(prdt_summary['Product'], prdt_summary['Sales'])
plt.title('TOTAL SALES BY PRODUCT')
plt.xlabel('PRODUCT')
plt.ylabel('TOTAL SALES')
plt.show()
pivot_table = df.pivot_table(values='Sales', index='Region', columns='Product',
aggfunc=np.sum, fill_value=0)
print(pivot_table)
correlation_matrix = df[['Sales','Quantity']].corr()
print(correlation_matrix)
import seaborn as sns
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

OUTPUT:

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call
drive.mount("/content/drive", force_remount=True).
        Date    Product  Sales  Quantity Region
0  01-01-2023  Product A    200         4  North
1  02-01-2023  Product B    150         3  South
```

2 03-01-2023 Product A 220 5 North

3 04-01-2023 Product C 300 6 East

4 05-01-2023 Product B 180 4 West

Date 0

Product 0

Sales 0

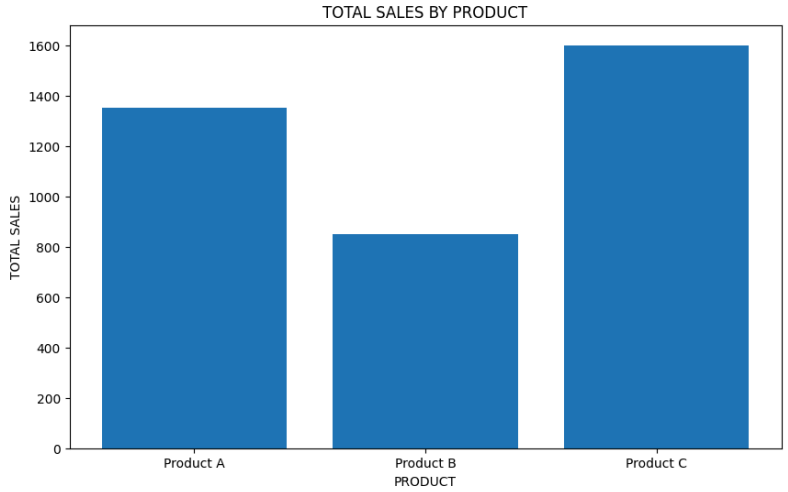
Quantity 0

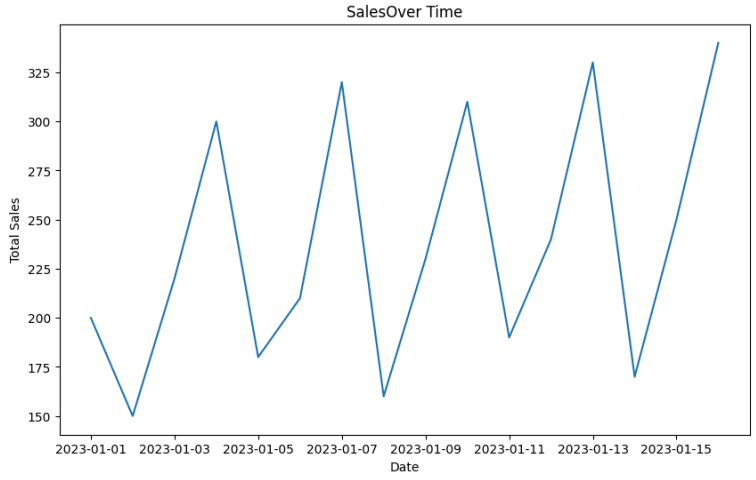
Region 0

dtype: int64

	Sales	Quantity
count	16.000000	16.000000
mean	237.500000	5.375000
std	64.031242	1.746425
min	150.000000	3.000000
25%	187.500000	4.000000
50%	225.000000	5.500000
75%	302.500000	7.000000
max	340.000000	8.000000

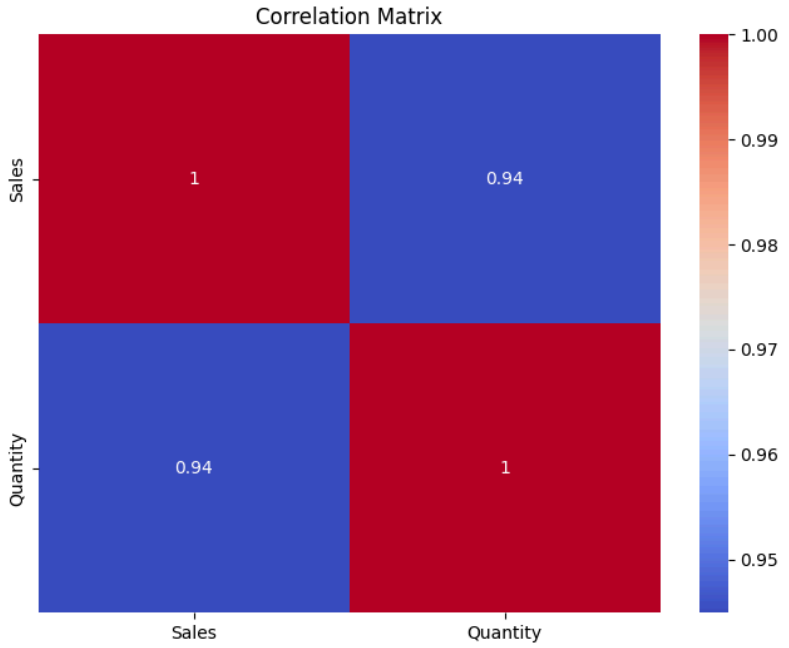
	Product	Sales	Quantity
0	Product A	1350	33
1	Product B	850	17
2	Product C	1600	36





Product	Product A	Product B	Product C
Region			
East	0	0	1600
North	1350	0	0
South	0	480	0
West	0	370	0

	Sales	Quantity
Sales	1.000000	0.944922
Quantity	0.944922	1.000000



RESULT:

Hence the data set given in csv format is uploaded and analyzed, and data preprocessing and visualization are performed.